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Review

Conceptual Architecture of a Trustworthy Wind and Photovoltaic Power Forecasting System: A Systematic Review and Design

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Abstract

Accurate and trustworthy forecasting of wind and photovoltaic power generation is essential for the reliable operation and planning of modern power systems. Although recent machine-learning-based forecasting solutions increasingly incorporate elements of trustworthy artificial intelligence, such as explainability, uncertainty quantification, robustness, drift monitoring, and machine learning operations, these components are typically introduced in a fragmented manner and remain weakly integrated at the architectural level, which limits their applicability in real operational environments. This paper presents a systematic review of 59 peer-reviewed journal articles published between 2019 and 2025, conducted in accordance with the PRISMA 2020 guidelines. The review includes studies focused on wind and photovoltaic power forecasting that report system architectures, frameworks, or end-to-end pipelines incorporating at least one trust-related attribute. The literature search was performed using Scopus, IEEE Xplore, MDPI, and ScienceDirect. Using a narrative and architectural synthesis, the review identifies six structural gaps hindering industrial deployment: the absence of semantic data models, shallow model-centric explainability, drift monitoring without governance mechanisms, lack of automated model lifecycle management, insufficient robustness to real-world data defects, and the absence of integrated end-to-end architectures. The evidence base is limited by the heterogeneity of architectural descriptions and the predominantly qualitative nature of reported implementations. Based on these findings, a high-level reference architecture for a trustworthy AI-based forecasting system is proposed. The architecture formalizes trustworthiness as a system-level property and integrates semantic, technological, and functional trust layers within a unified data and model lifecycle, supporting reproducible, interpretable, and operationally reliable forecasting for both wind and photovoltaic power plants.

Keywords: trustworthy AI; wind power forecasting; photovoltaic forecasting; explainable AI; uncertainty quantification; MLOps; drift monitoring; semantic modeling

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1. Introduction

Forecasting the power generation of renewable energy sources, particularly wind and photovoltaic (PV) systems, plays a critical role in modern power systems by enabling grid operators to plan operating regimes in advance and maintain power balance. However, conventional machine learning models used for such forecasting tasks often operate as "black boxes," which hinders understanding of their decision-making processes and reduces user trust in the resulting predictions. In recent years, the concept of trustworthy artificial

intelligence (trustworthy AI) has emerged, aiming to improve the transparency, robustness, and reliability of predictive models.

Key components of trustworthy AI include explainable artificial intelligence (XAI), which provides insights into model behavior and supports informed interpretation of forecasts; uncertainty quantification (UQ), for example through prediction intervals, which enables assessment of forecast confidence; data and model drift monitoring to ensure timely adaptation to changes in input data distributions; as well as machine learning operations (MLOps) practices and model governance mechanisms that support reliable deployment, monitoring, and maintenance of models in industrial environments. Collectively, these aspects are intended to ensure that renewable energy forecasts are not only accurate, but also transparent, robust to data degradation and external disturbances, and operationally reliable.

This paper presents a systematic review of recent studies published between 2019 and 2025 in which these trust attributes are incorporated into the architectures of wind and photovoltaic power forecasting systems. The review identifies key methods and architectural solutions aimed at enhancing trust, ranging from XAI-based model interpretation to concept drift monitoring and the adoption of MLOps pipelines. Building on the outcomes of the review, a conceptual architecture for renewable energy generation forecasting is proposed, synthesizing best practices reported in the literature. The proposed architecture is designed to improve model transparency for experts and operators, enhance robustness to data quality degradation and external impacts, and support a continuous model improvement cycle under real operational conditions.

In line with the objectives of this study and the identified need to systematize contemporary approaches to trustworthy AI for wind and PV power forecasting, the research aims to address the following research questions:

RQ1: Which trust attributes—such as explainability, uncertainty quantification, robustness, data and model drift monitoring, MLOps/model governance, and architectural consistency—are represented in current studies on wind and photovoltaic power forecasting?

RQ2: What structural gaps, limitations, and unresolved challenges exist in contemporary architectures for renewable energy forecasting from the perspective of trustworthy AI?

RQ3: What requirements should be imposed on a forecasting system to ensure semantic, technological, and functional trustworthiness under real operational conditions of generation assets?

RQ4: How can the results of the systematic review and the identified structural gaps be synthesized into a unified high-level architecture of a trustworthy AI system suitable for operational and short-term forecasting of wind and photovoltaic power generation?

The main contributions of this paper can be summarized as follows:

1. A systematic review of architectures for trustworthy AI in wind and photovoltaic power forecasting is conducted, with a focus on architectural rather than algorithmic gaps.
2. The concept of three trust layers—semantic, technological, and functional—is formalized in the context of renewable energy forecasting.
3. A high-level end-to-end reference architecture is proposed that addresses the identified structural gaps.
4. A traceability framework is introduced that maps trust attributes to architectural modules and associated measurable artifacts.

The remainder of the paper is organized as follows. Section 2 describes the literature selection methodology based on the PRISMA 2020 protocol, including the applied

inclusion and exclusion criteria. Section 3 presents a systematic review of trustworthy AI methods for wind and photovoltaic power forecasting and analyzes the key structural gaps identified in the existing literature. In Section 4, a high-level architecture of a trustworthy forecasting system is proposed, integrating semantic, technological, and functional trust layers. Section 5 discusses the results by relating the proposed architecture to the identified gaps and examining its applicability under real operational conditions. Finally, Section 6 summarizes the main conclusions of the study and outlines directions for future research.

2. Article Selection Criteria and Methodology

This systematic literature review (SLR) was conducted and reported in accordance with the PRISMA 2020 guidelines [1]. The primary objective of the review is to identify and analyze recent scientific publications addressing the design of trustworthy AI system architectures for renewable energy generation forecasting, with a particular focus on PV and wind power.

2.1. Data Sources

The literature search was conducted using the following indexed bibliographic databases: Scopus, IEEE Xplore, MDPI, and ScienceDirect. In addition, open repositories (e.g., ResearchGate) and previously published systematic reviews were used as secondary meta-sources to identify potentially relevant studies.

The search was restricted to peer-reviewed journal articles classified as Article or Review and published between 2019 and 2025. Conference proceedings were not included in the initial search but could be considered if the number of relevant journal publications proved insufficient.

To ensure stable coverage of the literature, transparent duplicate removal, and reproducibility of the search process, curated bibliographic databases (such as Scopus and IEEE Xplore) were used as primary sources. Web of Science was not used as a primary source due to its substantial overlap with Scopus for the considered research domain and time period. Databases such as Google Scholar were not employed as primary search sources, as they are not curated and aggregate journal articles, conference papers, and preprints with non-peer-reviewed materials, which may reduce the verifiability and methodological rigor of the SLR protocol.

The last search was conducted in November 2025.

2.2. Search Strategy and Keywords

To identify relevant studies, a search strategy based on multiple independent sets of search queries was employed. The queries were executed separately and were designed to capture combinations of the following key aspects: (1) forecasting methods, (2) application domain (wind or photovoltaic power generation), (3) trust attributes (e.g., explainable AI, data or concept drift, MLOps), and (4) system architecture or framework design. The general structure of the search queries is presented below:

- (“explainable ai” OR “xai”) AND (“forecasting”) AND (“wind power”) AND (“system architecture” OR framework OR “system design”);
- (“explainable ai” OR “XAI”) AND (“forecasting”) AND (“photovoltaic” OR “solar power”) AND (“system architecture” OR “framework” OR “system design”);
- (“concept drift” OR “data drift”) AND (“forecasting”) AND (“wind” OR “photovoltaic” OR “solar”) AND (“architecture” OR “framework” OR “pipeline”);
- (“MLOps” OR “model governance”) AND (“forecasting”) AND (“wind” OR “photovoltaic” OR “solar”) AND (“system architecture” OR “framework” OR “pipeline”);

- ("MLOps" OR "model governance") AND ("forecasting") AND ("wind" OR "photovoltaic" OR "solar") AND ("architecture" OR "framework" OR "pipeline");
- ("interpretable machine learning") AND forecasting AND ("wind power") AND ("system architecture" OR framework OR "system design");
- ("interpretable machine learning") AND forecasting AND (photovoltaic OR "solar power") AND ("system architecture" OR framework OR "system design");
- ("interpretable machine learning") AND forecasting AND (wind OR photovoltaic OR solar) AND (architecture OR framework OR pipeline);
- ("feature importance") AND forecasting AND ("wind power") AND ("system architecture" OR framework OR "system design");
- ("feature importance") AND forecasting AND (photovoltaic OR "solar power") AND ("system architecture" OR framework OR "system design");
- ("feature importance") AND forecasting AND (wind OR photovoltaic OR solar) AND (architecture OR framework OR pipeline).

The temporal scope of the review was restricted to the period 2019–2025 in order to capture the phase of active development and dissemination of trustworthy AI concepts—such as explainable models, data and concept drift monitoring, and MLOps practices—within the domain of renewable energy forecasting. This time-frame selection ensures methodological consistency and comparability across the analyzed studies.

2.3. Inclusion and Exclusion Criteria

Studies were included in the review if they met the following criteria:

- They addressed forecasting of wind or photovoltaic power generation;
- They described a system architecture, framework, system design, or processing pipeline;
- They considered at least one trust attribute, including XAI, UQ, data or concept drift monitoring, MLOps practices, robustness, or model governance;
- They were published in peer-reviewed journals, including journal preprints.
- Studies were excluded from the review if they:
- Had no explicit relevance to wind or photovoltaic power generation forecasting;
- Did not provide a description of a system architecture, framework, system design, or pipeline;
- Did not incorporate any trust-related attributes (XAI, UQ, drift monitoring, MLOps, robustness, or governance).

In addition, selected publications providing background on wind and photovoltaic power forecasting methods and their historical development were used to support contextual understanding of the domain.

The study selection process was performed by two reviewers (P.V.M., I.F.I) independently. Disagreements were resolved through discussion until consensus was reached.

At each stage of the selection process—identification, screening, eligibility assessment, and final inclusion—the number of studies and the reasons for exclusion were recorded. The results of the selection process are summarized in Figure 1.

2.4. Results of the Selection Process

In accordance with the PRISMA 2020 guidelines, a total of 282 publications were initially identified across the selected databases. After removing 82 duplicate records and one non-English language record, 199 studies remained for title and abstract screening. At this stage, 131 studies were excluded. Subsequently, 68 full-text articles were assessed for eligibility, all of which were accessible. During the eligibility assessment, 9 studies were excluded: 8 for not containing wind or photovoltaic power generation forecasts, and 1 for

not incorporating relevant trust-related attributes. As a result, 59 studies were included in the final review. A detailed breakdown of the selection process is presented in Figure 1.

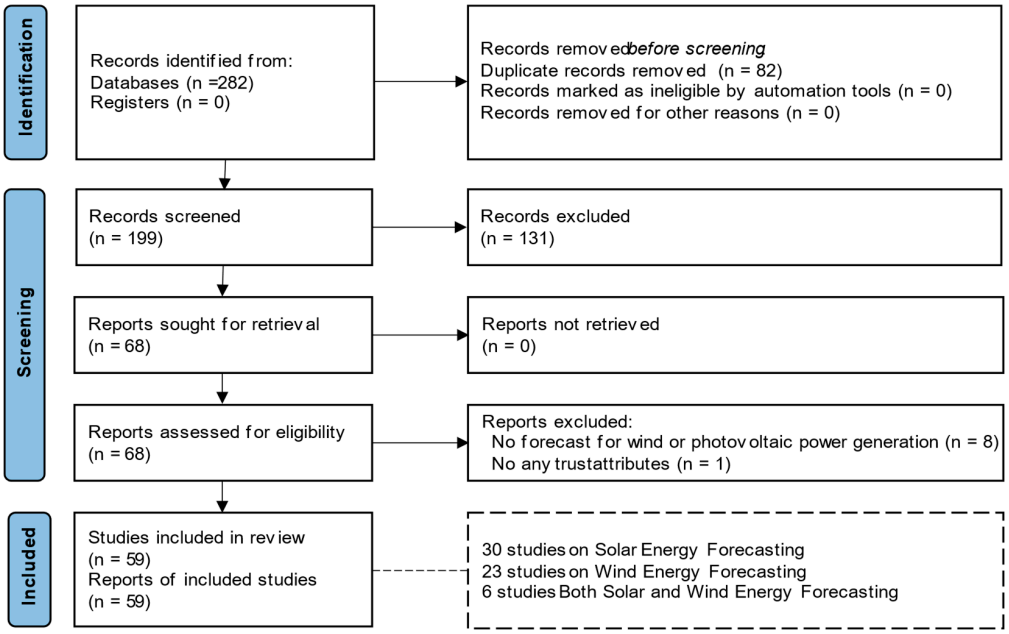


Figure 1. PRISMA 2020 flow diagram illustrating the identification, screening, eligibility assessment, and inclusion of studies in the systematic review of trustworthy wind and photovoltaic power forecasting architectures.

The included studies were distributed as follows (Figure 2a):

- Thirty studies focused on photovoltaic power forecasting;
- Twenty-three studies addressed wind power forecasting;
- Six studies considered both wind and photovoltaic power forecasting.

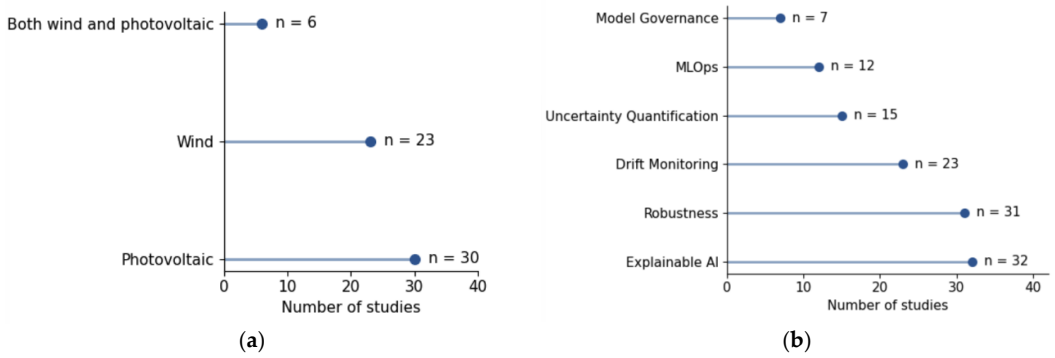


Figure 2. Distribution of studies by focus (**a**) and trust attributes (**b**) in the reviewed literature.

The analysis further revealed the distribution of trust attributes across the selected studies (Figure 2b):

- Thirty-two studies incorporated explainable artificial intelligence;
- Thirty-one studies addressed robustness;
- Twenty-three studies included data or concept drift monitoring;
- Fifteen studies implemented uncertainty quantification;
- Twelve studies considered MLOps practices;
- Seven studies incorporated model governance mechanisms.
- The most frequently observed combinations of trust attributes were:
- XAI and robustness;

- Drift monitoring and robustness;
- XAI and drift monitoring.

2.5. Limitations

Despite the comprehensive scope of the analysis, this study has several limitations. The review primarily focuses on the methodological and architectural principles underlying trustworthy renewable energy forecasting systems and does not include a quantitative validation of the proposed architecture using real operational data. In addition, the examined approaches to explainability, UQ, and drift monitoring are, to a considerable extent, dependent on specific implementation choices and hyperparameter settings, which may limit the generalizability of the findings.

Furthermore, the scope of the study does not encompass a detailed analysis of computational complexity or the associated operational costs related to the deployment and maintenance of the proposed pipeline in industrial environments.

2.6. Review Registration and Protocol

This systematic literature review was retrospectively registered in the Open Science Framework (OSF), registration ID: 84h6q. The registration is currently under embargo to preserve the blind peer-review process and will be made publicly available upon completion of the review process. No formal prospective review protocol was prepared.

3. Analysis of Trustworthy Renewable Energy Forecasting Approaches

Based on the selected 59 sources, a thematic classification of existing approaches to the design of trustworthy renewable energy forecasting systems was conducted. Each subsection below focuses on a specific trust attribute integrated into contemporary systems, including explainability, drift monitoring, UQ, robustness, MLOps, and model governance. For each attribute, representative examples from the literature are discussed, illustrating the applied methods, achieved effects, and key findings, supported by summarized evidence presented in tables.

3.1. Explainable AI in Energy Forecasting

Model explainability is a key component of trust, as it enables users and domain experts to understand why a model produces a particular forecast. Table 1 presents a structured analysis of key studies in which XAI methods are integrated into model architectures for wind and photovoltaic power generation forecasting. The analysis highlights the role of XAI in addressing critical challenges, including robustness to cyber-attacks and data drift, monitoring of concept shift, and quantitative assessment of forecast uncertainty.

Table 1. Overview of studies integrating XAI into wind and photovoltaic power forecasting architectures.

Reference	Authors	Tile	Architectures and Trust Attributes
[2]	A. Sundararajan; M. Olama; M. Ferrari; B. Ollis; G. Liu	A Data Quality-Aware Framework to Reliably Forecast Photovoltaic Generation and Consumer Load for an Improved Resilience of Microgrids	Data quality-aware framework incorporating data and concept drift monitoring and addressing model robustness under data quality degradation.
[3]	D. Astolfi; F. De Caro; A. Vaccaro	Wind Power Applications of eXplainable Artificial Intelligence Techniques	HBOLA (Hedge Backpropagation-based Online Long Short-Term Memory Architecture) with concept drift monitoring, uncertainty quantification via probabilistic forecasting, and mechanisms to enhance model robustness.

Table 1. *Cont.*

Reference	Authors	Tile	Architectures and Trust Attributes
[4]	V. K. Saini; A. S. Al-Sumaiti; R. Kumar	Integrating Data Intrusion Defense Strategy in PV Power Forecasting Framework	Framework for PV power forecasting protection against cyber-attacks, integrating robustness with adversarial disturbances, XAI as a tool for attack detection, and elements of model governance and MLOps.
[5]	D. Akpuluma; W. -G. Fruh; N. Firoz	Enhancing Solar Power Forecasting in Abuja: An Application of XAI Techniques for Improved Transparency and Trust	1D-CNN-based architecture for time series forecasting incorporating XAI, including two state-of-the-art methods for model interpretation.
[6]	M. Alagarsamy; U. Rajasekaran; S. Ganesan; R. Palanivel	XAI-Based Photovoltaic Energy Management Framework for Smart Cities	XPEMM framework (XAI-based Photovoltaic Energy Management Model chain) employing XAI through the LIME method and addressing robustness with respect to error accumulation.
[7]	Syed Ahmed Kabir, Ijaz F.; Gajendran, Mohan K.; Taslim, Prajna M.; Boopathy, Sethu R.; Ng, Eddie Y.; Mehdizadeh, Amirfarhang	An XAI Framework for Predicting Wind Turbine Power under Rainy Conditions Developed Using CFD Simulations	Pipeline combining CFD-based data generation, XAI model development, and integration into the Blade Element Momentum (BEM) method for wind turbine power estimation under rainy conditions, with explainable AI implemented via symbolic regression.
[8]	Zhang, L.; Zhu, J.; Zhang, D.; Liu, Y.	An incremental photovoltaic power prediction method considering concept drift and privacy protection	Comprehensive forecasting architecture (pipeline) including offline and online phases, federated learning (FL), incremental learning, concept drift monitoring, and privacy preservation mechanisms.
[9]	Li, J.; Jia, L.; Zhou, C.; Zhu, B.	Probability Density Function Control-Based Deep Ensemble Learning for Wind Energy System Power Forecasting	Three-stage forecasting pipeline consisting of data decomposition, prediction, and ensemble learning, with probability density function (PDF) control for error distribution management.
[10]	Konstantinou, G.; Hatziargyriou, N	Complex Terrains and Wind Power: Enhancing Forecasting Accuracy through CNNs and DeepSHAP Analysis	CNN-based wind power forecasting approach for regions with complex terrain; DeepSHAP to interpret feature contributions.
[11]	A. Meng; H. Liu; L. Xiao; Z. Tan; Z. Huang; Q. Zhang; B. Yan; H. Yin	An Explainable Multi-Feature Dimensionality Reduction Framework: Considering Trend Inconsistency in Wind Power Sample	Explainable dimensionality reduction framework that compensates dynamic feature–power trend inconsistencies and mitigates abnormal data effects; Shapley Additive exPlanations (SHAP) based global feature importance analysis.
[12]	M. Massaoudi; I. Chihi; L. Sidhom; M. Trabelsi; S.S. Refaat; F.S. Oueslati	Enhanced Random Forest Model for Robust Short-Term Photovoltaic Power Forecasting Using Weather Measurements	Random Forest–based photovoltaic power forecasting model enhanced with ensemble feature importance techniques; trust attributes addressed at the model level.
[13]	Y. Zhao; Y. Zhao; H. Liao; S. Pan; Y. Zheng	Interpreting LASSO Regression Model by Feature Space Matching Analysis for Spatio-Temporal Correlation-Based Wind Power Forecasting	Self-interpreting LASSO-based wind power forecasting framework; addresses explainability through white-box model interpretation.

The application of XAI techniques increases operator confidence by enhancing decision transparency and allowing the rationale behind predictions to be examined. Pan et al. [14] proposed a specialized architecture termed HBOLA for wind farm power

forecasting, incorporating an XAI module based on SHAP to interpret neural network outputs. DeepSHAP-based explainability has also been applied to analyze spatial and meteorological drivers of wind power generation in complex terrain, enabling physically interpretable insights that extend conventional feature importance analysis [10].

Astolfi et al. [3] introduced an interpretable framework based on sequential feature selection and Shapley values to address two key wind energy tasks: condition monitoring through a multidimensional power curve and ultra-short-term forecasting. The authors demonstrated that operational variables such as the standard deviation of generator speed and blade pitch angle can be more informative for the model than certain environmental factors.

The use of XAI techniques—such as Local Interpretable Model-agnostic Explanations (LIME), SHAP, and symbolic regression—has also been reported in a number of other studies for both wind power data [7,11,15–17] and solar power data [4,5,12,15,16,18,19]. For example, Kabir et al. [7] developed a specialized XAI framework based on symbolic regression to estimate wind turbine power under rainy conditions. This approach enabled faster computations compared to classical CFD-based models and yielded interpretable relationships between rainfall intensity and turbine aerodynamics. An example of explainable dimensionality reduction for wind power forecasting is provided in the paper of Meng et al. [11], where trend-consistency–guided feature selection is interpreted using SHAP to preserve physically meaningful feature–power relationships.

Alagarsamy et al. [18] and Jian et al. [20] applied LIME and attention mechanisms within solar energy forecasting models to visualize the importance of different time steps and input features. Furthermore, the Liao et al. [21] proposed quantitative metrics to assess the reliability of interpretations produced by SHAP, Permutation Feature Importance, Partial Dependence Plots, and LIME in the context of wind power forecasting.

Beyond post hoc explainability of black-box models, self-interpreting regression approaches have also been explored, where the decision logic of LASSO-based wind power forecasting models is elucidated through feature space matching and spatio-temporal correlation analysis [13].

Interpretable feature selection approaches have also been proposed for solar irradiance forecasting, where feature relevance is assessed through perturbation- and variance-based importance measures [22].

Across nearly all reviewed studies, XAI serves not only as a tool for improving transparency but also as a mechanism for model validation. The analysis of explanations helps identify non-informative features and potential modeling errors, which can ultimately lead to improved forecasting accuracy [4,5,20,23].

Table 2 summarizes the characteristics of selected examples of XAI integration. Overall, the inclusion of XAI modules in renewable energy forecasting systems contributes to increased user trust and facilitates the adoption of models in operational practice, as it makes forecasts more interpretable and verifiable by domain experts. Representative examples of XAI applications in wind and photovoltaic power forecasting are presented in Table 2.

Table 2. XAI application example for RES forecasting.

Reference	Solar/Wind Power Plant	XAI Methods	XAI Usage
[5,20]	Solar	Permutation Feature Importance, Integrated Gradients	Post hoc analysis of feature importance and their temporal dynamics
[24]	Solar	SHAP, LIME	Global and local interpretation of an ensemble forecasting model

Table 2. *Cont.*

Reference	Solar/Wind Power Plant	XAI Methods	XAI Usage
[7]	Wind	Symbolic regression	Development of an interpretable physics-informed model to characterize the impact of rainfall
[4]	Solar	Shannon Entropy, Kurtosis	Feature extraction for a cyber-attack detection module
[25]	Solar	SHAP	Automated selection and interpretation of the optimal machine learning model (AutoML) for photovoltaic power forecasting, enabling global explainability
[26]	Solar	SHAP, LIME, Anchors, PDP	User-oriented framework designed to ensure transparency and trust in AI-based energy services (explainability by design)
[27]	Solar	SHAP, LIME, Anchors, PDP	Comprehensive framework (El-Build) with user-adaptive explanation mechanisms
[10]	Wind	DeepSHAP	Spatial and meteorological driver attribution for wind power forecasting in complex terrain; physically interpretable explanation of model behavior beyond standard feature importance
[13]	Wind	Feature space matching analysis; sensitivity and perturbation analysis	White-box interpretation of a self-interpreting regression model; feature selection under spatio-temporal correlations and assessment of interpretation consistency across settings

3.2. Monitoring of Concept Drift and Data Drift

Concept drift and the related phenomenon of data drift refer to changes in data distributions or in the relationship between input features and the target variable over time, which can significantly degrade the performance of models trained on historical data. In the context of renewable energy forecasting, drift may arise due to changes in climatic conditions, turbine upgrades, equipment degradation, or the emergence of new external influencing factors. Such changes may occur gradually, abruptly, or in a recurring manner, and all of these patterns are commonly observed in energy generation time series. Consequently, a number of studies have focused on automated drift detection and adaptive model updating mechanisms [8,14,28–30].

Zhang et al. [8] proposed an incremental forecasting approach for photovoltaic power generation that accounts for concept drift while preserving data privacy. Their comprehensive pipeline integrates offline and online training phases, federated learning for data protection, and a drift monitoring module that triggers model retraining when distributional shifts are detected. In the paper of Pan et al. [14], an architecture for wind power forecasting also incorporates online drift detection, combined with probabilistic forecasting to support uncertainty assessment. Luo et al. [30] introduced an Adaptive Long Shot-term Memory (AD-LSTM) framework with a two-phase learning strategy (TP-ALS), enabling switching between stationary and adaptive training phases upon drift detection using the SDWIN algorithm. Furthermore, Lyu et al. [31] presents an Online NeuroEvolution-based Neural Architecture Search (ONE-NAS) algorithm for the automatic design and dynamic online training of recurrent neural networks, allowing adaptation to non-stationary data characteristic of wind energy applications.

Examples of approaches for drift detection and handling are summarized in Table 3.

Table 3. Methods for drift monitoring and adaptation.

Reference	Drift Type	Methods	Results
[32]	Concept	Mutual information and Minkowski distance–based drift detection within the FEEL framework	Improvement in R^2 from approximately 48.3% to 8.1% variance dispersion, indicating enhanced generalization performance by a factor of two.
[33]	Concept	Enhanced TimesNetV2 combined with Inception V2 for drift detection, and PS-HBPMamba for model adaptation	Reduction in root mean square errror (RMSE) by 36.93%, mean absolute error (MAE) by 41.26%, and SMAPE by 56.57%, with an associated R^2 improvement of 1.66%.
[30]	Concept	SDWIN algorithm for drift detection and Adaptive LSTM (AD-LSTM) with a two-phase adaptive learning strategy (TP-ALS).	Forecasting accuracy improved to 73.11%.
[34]	Data	ONE-NAS (online neural architecture search) for adaptive model design and training	Performance exceeding that of conventional baseline methods, including offline Autoregressive Integrated Moving Average (ARIMA), LSTM, and GRU models.

Timely adaptation of forecasting models to drift—through continuous monitoring of prediction error statistics, deviations in input data distributions, and related indicators—significantly enhances the robustness of forecasting systems during long-term operation [35]. Consequently, it is recommended that forecasting system architectures incorporate dedicated components for continuous monitoring of both prediction quality and incoming data characteristics, enabling automated model updates when necessary. In the absence of such mechanisms, even highly accurate algorithms tend to degrade over time due to shifts in input data distributions, a phenomenon commonly referred to as model aging.

3.3. Uncertainty Quantification in Forecasting

In addition to point accuracy, uncertainty plays a critical role in energy forecasting. Energy companies value not only point estimates of power generation but also confidence intervals or probabilistic assessments that enable evaluation of the risk associated with deviations between predicted and actual generation. As a result, probabilistic forecasting methods are increasingly applied to wind and photovoltaic power forecasting [9,14,36–39]. For example, the construction of prediction intervals (PIs) around point forecasts provides an effective means of representing uncertainty and improving the reliability of decision-making by grid operators [10,32,36,39–41].

Uncertainty quantification methods are also applied beyond interval estimation and are increasingly used in operational power system management tasks. Esmaeel Nezhad et al. [6] applied chance-constrained optimization for energy hub scheduling, enabling quantitative consideration of risks arising from forecasting uncertainty in photovoltaic generation and load demand.

Across the reviewed studies, uncertainty quantification is implemented using a variety of approaches, including ensemble-based methods [5,12,19,32,41,42], error distribution modeling [37,43,44], quantile regression [10], among others. For instance,

Massaoudi et al. [12] combined ensemble-based learning with feature importance analysis to enhance interpretability and robustness of photovoltaic power forecasting models.

In the paper of Pan et al. [14], the HBOLA architecture employs a probabilistic LSTM-based approach that predicts a distribution of power output rather than a single value, thereby enabling the computation of confidence intervals and a quantitative assessment of forecast uncertainty. Li et al. [9] proposed a multi-stage wind forecasting model based on deep ensembles, incorporating probability density function (PDF) control. In this approach, the model is optimized to enforce predefined properties of the error distribution—such as reduced variance and the absence of heavy tails—resulting in high-quality prediction intervals. Mitrentsis et al. [45] presented a probabilistic photovoltaic power forecasting approach based on natural gradient boosting, with uncertainty evaluated through prediction intervals and continuous ranked probability score (CRPS).

Table 4 provides an overview of uncertainty quantification approaches applied in renewable energy forecasting. Overall, the reviewed studies consistently indicate that providing prediction intervals or other uncertainty metrics enhances trust among system operators and engineers by offering insight into potential deviations and enabling proactive reserve planning [46–48]. Moreover, explicit uncertainty information facilitates model comparison not only in terms of average error but also in terms of reliability; for example, one algorithm may produce narrower intervals at the same confidence level, indicating higher certainty and potentially superior generalization capability.

Table 4. Probabilistic and interval forecasting approaches.

Reference	Method	Uncertainty Evaluation Metrics
[32]	Ensemble-based methods	Probability Integral Transform (PIT), Expected Calibration Error (ECE)
[14]	Quantile regression (Quantile)	Coverage, ECE
[39]	Ensemble Model Output Statistics (EMOS)	Coverage, ECE
[9]	Ensemble-based methods	ECE
[45]	NGBoost probabilistic regression with prediction intervals	CRPS; PIT

3.4. Robustness and Cyber-Resilience of Forecasting Models

Robustness of forecasting models refers to their ability to withstand various adverse factors, including data outliers and missing values, malicious interventions (e.g., cyber-attacks), and extreme conditions for which the model was not explicitly trained. In industrial power systems, robustness is of critical importance, as forecasting failures may lead to economic losses or even emergency situations. One key aspect of robustness is data quality: contaminated or incomplete data can degrade the performance of even highly sophisticated models (the well-known "garbage in–garbage out" principle). Therefore, forecasting architectures must incorporate data quality control and preprocessing mechanisms. Sundararajan et al. [2] developed a Data Quality-Aware framework for photovoltaic power forecasting, which monitors input data quality metrics, detects anomalies, and thereby maintains forecasting accuracy even under data degradation.

Another critical aspect is protection against adversarial influences and cyber-attacks, which is particularly relevant in the context of federated learning and IoT-based systems [49]. Saini et al. [4] proposed an architecture for photovoltaic power forecasting with built-in cybersecurity mechanisms. The system includes an attack detection module that leverages XAI techniques to identify anomalous patterns, as well as dedicated procedures to preserve model functionality under attack conditions. The same study also reports the

use of model governance and MLOps elements that enable rapid replacement or updating of compromised models with validated versions.

Robustness to anomalous and trend-inconsistent data can also be addressed at the preprocessing or model level, for example through dimensionality reduction methods designed to compensate dynamic feature offsets or through explicit outlier-aware forecasting frameworks [11,19].

Robustness with respect to physical and environmental factors has also been investigated. Kabir et al. [7] evaluated the impact of heavy rainfall on wind turbine power forecasting accuracy and developed an XAI-based model to account for such extreme conditions, exhibiting only a modest accuracy degradation (approximately 5–7%) even under intense precipitation. Das et al. [37] applied machine learning techniques to improve forecast reliability and evaluate the performance of photovoltaic panels under extreme Antarctic conditions. The study of Benovic et al. [50] examines the role of solar energy integration and AI technologies in sustainable local development, including aspects related to system reliability. Furthermore, Kalra et al. [51] demonstrates an approach to enhancing the robustness of forecasting models in extreme environments, while [52] investigates the resilience of wind energy systems to changes in power system conditions.

Examples of solutions aimed at improving robustness are summarized in Table 5.

Table 5. Robustness-oriented forecasting solutions and mitigation measures.

Reference	Forecasting System	Sources of Accuracy Degradation	Implemented Mitigation Measures
[2]	PV power and load forecasting	Missing data, epistemic uncertainty, data and concept drift	Framework with missing data handling, discrepancy tests, performance monitoring, and dynamic switching between models
[4]	PV power forecasting	Cyber-attacks (data poisoning), outliers	XAI and t-SNE for feature analysis; CNN–LSTM with attention for robust forecasting; protection against poisoning attacks
[36]	Wind speed forecasting	High volatility and intermittency of wind	CNN–BiLSTM cascade; hyperparameter optimization via BOHB; feature selection using TMGWO
[24]	Wind power forecasting	Model interpretability challenges, nonlinear dependencies	CatBoost + XGBoost ensemble with residual learning; XAI-based interpretation
[53]	PV and wind power forecasting	Distributional uncertainty, concept drift	Evolutionary clustering; model adaptation to changes in streaming data
[43]	PV power forecasting	Weather-induced fluctuations	Recurrent neural networks (RNN), Gated recurrent units (GRU), LSTM with attention; hyperparameter optimization via modified Particle Swarm Optimization
[19]	PV power forecasting	Outliers/abnormal measurements	Outlier-aware pipeline combining outlier occurrence prediction and a dedicated forecasting model for abnormal regimes, improving robustness under anomalous PV measurements

The reviewed studies emphasize the necessity of multi-level protection for forecasting models, encompassing the data level (noise filtering and missing data imputation) [2,8], the model level (robust algorithms, ensemble methods, and error distribution control) [4,9,32], and the operational level (anomaly monitoring, fallback models, and cybersecurity mechanisms) [54–56]. The integration of XAI with reliability-enhancing techniques is considered particularly promising, as model interpretation can help identify unexpected failures and vulnerabilities, thereby improving overall system reliability [5,16,36].

3.5. MLOps Practices and Automation of Forecasting Pipelines

To transition forecasting models from research prototypes to industrial deployment, MLOps methodologies have become increasingly important. MLOps encompasses a set of principles and tools for the continuous integration, delivery, and maintenance of machine learning models. Although the adoption of MLOps in energy forecasting is still emerging, several successful implementations have already been reported. Oyucu et al. [56] developed an end-to-end MLOps pipeline for short-term wind power forecasting in Türkiye. The proposed platform automates the entire workflow—from supervisory control and data acquisition (SCADA) data processing and selection and training of optimal models (across multiple ML/DL algorithms) to containerization using Docker and deployment with orchestration via Docker Swarm. The reported results demonstrate that the introduction of MLOps significantly improved system performance, reducing forecast latency to 9 ms, while also enhancing operational reliability through containerization and version control, and improving scalability of the forecasting services.

In general, an MLOps architecture supports automation across model development, testing, deployment, and monitoring, enabling new model versions to progress rapidly and safely from experimental stages to production. Continuous monitoring of model performance in production environments is a core component of MLOps, including detection of drift and accuracy degradation, automated rollback to previous versions in case of failures, and systematic management of data and experiments. Zhao et al. [33] proposed a comprehensive MLOps-compatible forecasting framework, incorporating formal quality control stages and decision gates at each phase, along with full traceability of model artifacts, including data, code, and evaluation metrics. The authors demonstrate that such a modular end-to-end platform, integrating MLOps practices with domain expertise, improves both forecasting accuracy and reliability, while also enhancing compliance with regulatory requirements under real operational conditions. In addition, Liu et al. [44] provides a comprehensive review of multidimensional data processing and intelligent forecasting techniques for renewable energy, emphasizing the importance of end-to-end pipelines.

Table 6 summarizes representative applications of MLOps in energy forecasting projects.

Table 6. Applications of MLOps in energy forecasting.

Reference	Key MLOps Pipeline Components	Achieved Benefits
[54]	Automated machine learning (AutoML), resource management, integration with energy infrastructure	Reduced complexity of AI capability management; improved planning and monitoring of microgrids
[4]	Protection against cyber-attacks (data poisoning), outlier handling	XAI and t-SNE for feature analysis; CNN–LSTM with attention for robust forecasting; protection against poisoning attacks
[55]	Predictive analytics, digital twins, distributed control, protocol standardization	Improved operational efficiency, robustness, and scalability of microgrids
[44]	Intelligent optimization algorithms, model performance monitoring	Reduced forecasting errors (RMSE, MSE); improved model robustness
[56]	End-to-end MLOps-pipeline, Docker, hyper-parameters tuning	Accelerated deployment (latency of 9 ms); improved reliability and scalability of forecasting models
[57]	End-to-end MLOps pipeline, Docker-based deployment, hyperparameter optimization, latency monitoring	Improved data quality, reduced deployment risks, and enhanced reproducibility
[49]	Federated learning (FL), data protection, management of heterogeneous devices	Improved security and privacy, as well as increased robustness to attacks in IoT networks

The integration of MLOps practices—ranging from version control of models and data to continuous integration and continuous deployment (CI/CD) pipelines for machine

learning—is a necessary condition for the long-term operation of complex renewable energy forecasting models [55,58]. Such integration ensures result reproducibility, enables rapid model evolution as new data become available, and simplifies system maintenance in production environments [57].

3.6. Model Governance and Trust

The final, yet equally important, trust attribute is model governance, which encompasses the processes and policies governing the model lifecycle, including model validation, versioning and configuration management, compliance with regulatory requirements, and auditing of quality-related characteristics. In the context of trustworthy systems, model governance provides an additional layer of confidence by ensuring that deployed models have undergone the necessary validation procedures and that any changes are controlled and transparent to relevant stakeholders. Studies explicitly focusing on governance aspects in the context of renewable energy forecasting remain relatively limited, accounting for 9 out of the 52 included studies [4,18,33,44,49,54–57].

Governance considerations are more frequently discussed in conjunction with MLOps practices [56,57]. For example, Saini [4] notes that the proposed framework incorporates elements of model governance, which likely implies the existence of a model repository, logging of training processes, access control, and mechanisms for model replacement in response to security threats. In the framework described by Zhao et al. [33], decision gate governance is explicitly introduced in the form of formalized checkpoints that determine model readiness to progress to subsequent lifecycle stages, with particular emphasis on regulatory compliance and model traceability in energy systems. Similarly, Oyucu et al. [56] highlights the importance of protocol standardization and the development of sustainable business models for microgrid management.

In practice, the model governance process can be conceptualized as a sequence of control checkpoints preceding model deployment in operational environments. Table 7 illustrates the key governance stages, associated validation criteria, and responsible roles.

Thus, several common directions of model governance can be identified:

- Technical consistency—verification that a model satisfies predefined quality and stability metrics (e.g., prior to replacing a previous model version) [18,33];
- Documentation and reproducibility—each model is accompanied by comprehensive descriptions of data, methods, and parameters, enabling auditing and reproducibility [44,57];
- Regulatory compliance—in a number of jurisdictions, grid operators are subject to specific requirements for forecasting systems, and deployed models must comply with them (e.g., explainability requirements and prescribed reliability indicators) [4,49].

The adoption of governance practices directly increases the trust of stakeholders—such as regulators and company management—in AI-based forecasting solutions [59,60]. As noted by Yaacoub et al. [49] and Alaraj et al. [57], without appropriate model governance it is difficult to claim full trust in a model, particularly in safety-critical applications. Contemporary approaches, including CRISP-ML, Model Cards, and related frameworks, contribute to improved model management and deployment reliability [54,55], but require adaptation to the specific characteristics and constraints of the energy sector.

In the proposed architecture (see the following section), governance aspects are addressed through a centralized model registry, systematic logging of updates and validation procedures, and the introduction of trust-related metrics (e.g., a Trust Scorecard for each model) [18,56]. This provides an additional layer of confidence for system users.

Table 7. Model governance elements in energy forecasting.

Decision Gate	Key Criteria	Responsible Roles	Related Studies
1. Data validation	Input data quality, absence of drift (PSI), handling of anomalies and missing values	Data Engineer, Data Scientist	[2,16]—data quality and drift monitoring
2. Technical model validation	Model performance (RMSE, MAE) exceeding baseline, stability on the validation dataset	ML Engineer, Data Scientist	[44,56]—comparison with baseline models and metric thresholds
3. Explainability validation	Availability of explanations (SHAP/LIME), consistency with physical process understanding	Data Scientist, Domain Expert	[18,36]—use of XAI for validation of physical interpretability
4. Robustness assessment	Robustness to noise, attacks, and operation under edge conditions	Security Engineer, ML Engineer	[4,49]—cyber-resilience and robustness
5. Regulatory compliance	Compliance with industry standards and regulatory requirements	Compliance Officer, Product Owner	[33,57]—traceability and regulatory alignment
6. Finalization and documentation	Comprehensive model documentation (Model Card), versioning, deployment protocol	ML Engineer, Technical Writer	[55,57]—documentation and reproducibility

3.7. Summary of Identified Gaps

The systematic review indicates that existing wind and photovoltaic power forecasting systems have achieved substantial progress in terms of model accuracy; however, they remain fragmented with respect to trustworthiness at the architectural level. Based on the findings of Sections 3.1–3.6, six structural gaps are identified that hinder the development of operationally viable trustworthy renewable energy forecasting systems.

1. Absence of unified semantic control and interpretable data models. Most existing architectures rely primarily on ad hoc feature engineering and lack a semantic backbone that defines domain entities, attributes, units of measurement, and relationships. The absence of semantic consistency leads to duplicated concepts, difficulties in integrating heterogeneous data sources (e.g., numerical weather prediction (NWP), SCADA, ERP), and limited extensibility of forecasting pipelines. This stands in contrast to established energy industry practices and industrial standards, e.g., Common Information Model, (CIM)/International Electrotechnical Commission (IEC) 61970, which require strict interoperability.

2. Explainability remains shallow and model-centric. Existing studies predominantly employ SHAP or LIME as post hoc tools for estimating feature contributions; however, very few transform these attributions into domain-relevant, human-readable explanations. In most systems, explainability does not influence architectural decisions, and no study presents a fully traceable explanation chain—from raw input data to the final forecast.

3. Underdeveloped monitoring of model degradation and data drift. Despite the use of drift detectors (e.g., ADWIN, Page–Hinkley, CUSUM) in isolated studies, only a limited number of forecasting systems implement continuous monitoring pipelines aligned with operational requirements. Drift is often treated as an experimental artifact rather than a real industrial risk. The lack of a managed drift control process results in silent model degradation and delayed intervention, thereby reducing trust in operational environments.

4. Lack of automated adaptation and lifecycle management of models. Even when drift detection mechanisms are present, models are rarely updated according to structured adaptation strategies. None of the reviewed studies proposes an integrated system that simultaneously combines:

- Version control;
- Safe retraining procedures;
- Validation thresholds (deployment gates);
- Model integrity checks;
- Rollback mechanisms.

5. Insufficient robustness to incomplete, noisy, and semantically inconsistent data. Many studies rely on "clean" experimental datasets and do not consider sensor failures, missing SCADA data, corrupted meteorological files, or inconsistent temporal indices. Such assumptions contradict real operating conditions at renewable energy facilities and industrial sites.

6. Absence of an end-to-end trustworthy architecture integrating all trust attributes. In the reviewed literature, research efforts remain fragmented across individual directions, including:

- Accuracy improvement (ML/Deep learning models);
- Explainability;
- Uncertainty quantification;
- Drift detection;
- Feature selection;
- MLOps practices.

However, no study integrates all these attributes into a single, verifiable, semantically consistent, and operationally applicable architecture. This unmet need forms the basis for the high-level architecture proposed in Section 4, which synthesizes semantic, technological, and functional trust layers.

The identified gaps are subsequently used as design requirements for the architectural synthesis presented in Section 4.

4. Proposed High-Level Architecture of a Trustworthy Wind and Photovoltaic Power Forecasting System

The proposed architecture provides a conceptual foundation for the development of trustworthy operational and short-term forecasting systems for wind and photovoltaic power generation. It is grounded in the structural gaps identified in Section 3 and formalizes the requirements that must be satisfied to enable the transition from isolated machine learning models to operationally reliable forecasting systems.

The architecture is organized around three complementary trust layers:

1. Semantic trust—consistency of data, features, models, and explanations, achieved through a unified domain ontology, data harmonization rules, and structural unification of feature representations.
2. Technological trust—the quality and robustness of the forecasting process, including data cleaning, feature extraction, regime clustering, robust modeling, uncertainty quantification, and drift monitoring.
3. Functional trust—transparency, controllability, and operational readiness of the system, ensured through explainability mechanisms, model state monitoring, MLOps practices, and model lifecycle management.

The architecture is agnostic to specific machine learning algorithms; however, it establishes strict principles governing how models are integrated into a production-grade

renewable energy forecasting pipeline, which stages must be implemented as mandatory components, and which trust-related artifacts must be generated at each layer.

4.1. Design Objectives and Principles

The primary objective of the proposed architecture is to establish a reliable, verifiable, and reproducible forecasting framework that remains robust under operational conditions characteristic of wind and photovoltaic power plants. The architecture is designed to operate under the following constraints:

- High variability of input data (e.g., wind regimes, cloud cover dynamics, sensor degradation);
- Incomplete or noisy telemetry (e.g., missing SCADA records, synchronization errors);
- Long-term changes in equipment characteristics and climatic conditions;
- The need to provide interpretable explanations to support engineering and dispatch decision-making;
- Requirements for transparency, auditability, and controllability of models in production environments.

Based on the structural gaps identified in Section 3, the architecture aims to ensure the following key properties:

1. Semantic integrity—a unified interpretation of data (SCADA, meteorological inputs, forecasts, and explanations) enabled by an explicit ontological layer.
2. System-level interpretability—explanations that extend beyond feature importance to include regime selection, model structure, uncertainty estimation, and the rationale behind drift detection events.
3. Automatable adaptation—controlled model updating triggered by detected drift, data degradation, or transitions between operational regimes.
4. Robustness and fault tolerance—built-in mechanisms for handling noise, outliers, missing values, extreme operating conditions, and structural inconsistencies in data.
5. Operational reliability—integration of MLOps practices and model lifecycle management (model governance), including validation gates, logging, version control, audit trails, and secure deployment procedures.

These principles form the basis of the high-level trustworthy forecasting architecture presented in Section 4.2 and visualized in Figure 3.

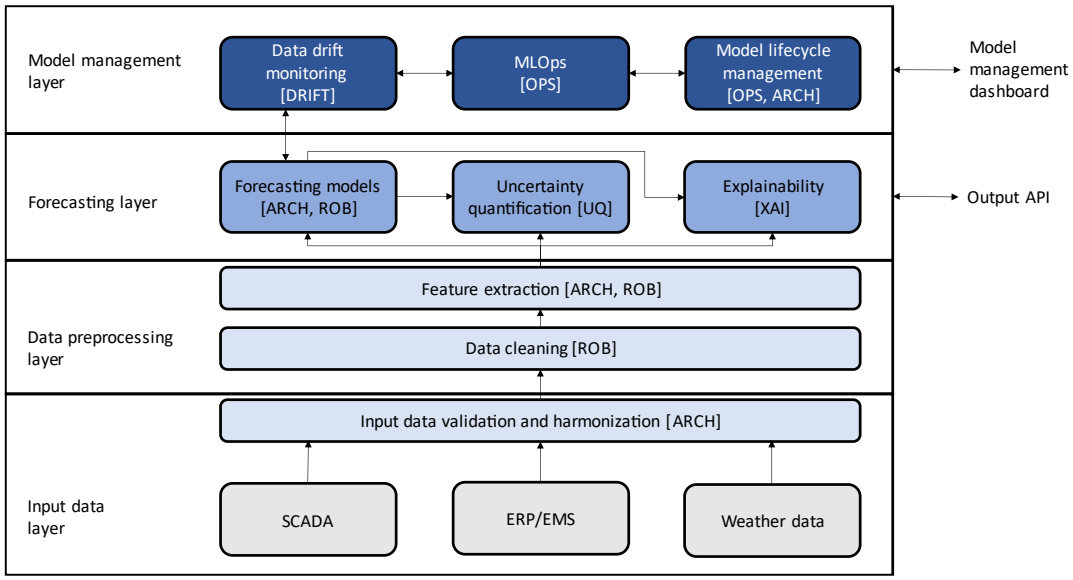


Figure 3. Conceptual architecture of the proposed system.

It is important to distinguish between the semantic layer and explainability mechanisms within the proposed architecture. The semantic layer serves as an infrastructural component responsible for harmonizing data, features, models, and domain concepts through an explicit ontological representation. It ensures consistent interpretation of physical quantities, operational states, units, and relationships across the entire forecasting pipeline.

Explainable AI, by contrast, operates at the level of model interpretation and decision transparency. XAI methods leverage the semantic layer to transform model attributions into domain-aligned, human-interpretable explanations, but do not replace the semantic layer itself. In this sense, semantics provide the structural substrate, while XAI constitutes a functional mechanism for interpreting model behavior and forecasting outcomes.

4.2. Overall Architecture Structure

The proposed architecture is designed as an end-to-end modular pipeline that integrates data processing, feature construction, forecasting, and operational support of predictive models. It is structured around the key trust attributes—semantic, technological, and functional—and ensures their continuous realization throughout all stages of the forecasting system lifecycle. In line with the structural gaps identified in Section 3, the architecture explicitly addresses the requirements for interpretability, robustness, adaptability, transparency, and controllability of forecasting models. A high-level representation of the system is illustrated in Figure 3.

The architecture is composed of four layers: the data ingestion layer, the preprocessing layer, the forecasting layer, and the model management layer. Each layer comprises one or more modules that implement specific trust attributes (ARCH, ROB, UQ, XAI, DRIFT, OPS—Table 8) and together form a coherent sequence of operations across the forecasting pipeline.

Table 8. Trust Attributes.

Marker	Full Name	Description
ARCH	Architectural Trust	Ensures structural integrity and consistency of the system, including domain ontologies, unified interfaces, standardized data schemas, and process reproducibility.
XAI	Explainability Trust	Guarantees transparency and interpretability of models through SHAP/LIME, ontological grouping of features, explanation graphs, and stability assessment of interpretations.
UQ	Uncertainty Quantification Trust	Ensures correct estimation of uncertainty using quantile regression, interval forecasts, CRPS, reliability metrics, PIT, and calibration mechanisms.
DRIFT	Drift Monitoring and Adaptation Trust	Addresses monitoring of data and model drift, detection of performance degradation, and implementation of safe model adaptation and update procedures.
OPS	Operational Trust (MLOps and Governance)	Ensures operational reliability through model registries, CI/CD pipelines, service monitoring, audit logs, rollback mechanisms, version control, and compliance with regulatory requirements.
ROB	Robustness Trust	Characterizes system resilience to missing data, noise, anomalies, and extreme scenarios, ensuring correct operation under degraded data quality conditions.

1. Data Ingestion Layer: Collection, Harmonization, and Semantic Alignment (ARCH). At the lowest level, the architecture includes an input data validation and harmonization module that ingests heterogeneous and asynchronous data sources. These sources comprise SCADA telemetry from wind turbines and photovoltaic inverters, data from enterprise resource planning (ERP), enterprise management system (EMS), meteorological forecasts (e.g., ERA5, ECMWF, WRF), as well as signals related to power curtailment, scheduled maintenance, equipment icing, and other operational states. The incoming data streams differ in sampling frequency, structure, units of measurement, and completeness.

The primary function of this module is to ensure semantic consistency by converting data into a unified format, validating data types and units, resolving structural inconsistencies, handling format-related errors, and synchronizing time series. At this stage, metadata and a common data schema are generated for use by downstream modules. In this way, the ARCH trust attribute is established, ensuring architectural consistency and reproducibility of the system.

2. Preprocessing Layer: Data Cleaning and Feature Extraction (ROB, ARCH). The preprocessing layer implements technological trust by ensuring robustness to data defects and by preparing features in a consistent and reliable manner.

2.1. Data Cleaning (ROB). The data cleaning module detects and corrects missing values, short-term outliers, noise distortions, and invalid SCADA telemetry readings. It incorporates algorithms for missing data imputation, noise filtering, temporal alignment, and the generation of data quality reports (DQ reports), which are used to monitor the reliability of incoming data streams and to support decision-making within MLOps processes. This stage is critical for trustworthiness, as it prevents data-related errors from propagating upward through the pipeline and ensures system resilience to operational anomalies that are inevitable in wind and photovoltaic power plants.

2.2. Feature Extraction and Regime Identification (ARCH, ROB). Following data cleaning, the system constructs features based on a predefined set of semantic rules and domain knowledge. Both basic and aggregated features are generated, including wind profiles, temperature indicators, solar incidence angles, cloudiness indices, power curtailment states, and others. A key component of this module is regime clustering, performed using SCADA and meteorological data. Characteristic wind regimes (e.g., calm conditions, gusty winds, nocturnal drops), solar generation regimes (e.g., clear sky, variable cloud cover, snow or icing), and operational regimes of the power plant are identified. The presence of such regime clusters enhances the robustness of subsequent models by enabling forecasts that explicitly account for regime-specific behavior.

As a result, this layer ensures not only robustness but also semantic coherence of the feature space.

3. Forecasting Layer: Models, Uncertainty, and Explainability (ARCH, ROB, UQ, XAI). The forecasting layer constitutes the core of the architecture and integrates the model ensemble, uncertainty quantification module, and explainability module.

3.1. Forecasting Models (ARCH, ROB). The architecture supports a wide range of forecasting models, including ensembles of decision trees (e.g., Gradient Boosting, LightGBM, CatBoost), recurrent and transformer-based neural networks, hybrid approaches, and models specialized for specific operational regimes. The architectural requirement is not the selection of a particular algorithm, but the standardization of how models are integrated into the production pipeline. Each model must exist as a versioned artifact accompanied by metadata, evaluation metrics, training provenance, and mechanisms for safe updating. This ensures both architectural and technological trust (ARCH, ROB).

3.2. Uncertainty Quantification (UQ). The UQ module performs probabilistic forecasting by generating prediction intervals or full predictive distributions. It employs

methods such as quantile regression, ensemble dispersion, EMOS calibration, and conformal prediction to assess interval overlap and calibration quality. Metrics such as CRPS, coverage, and PIT serve as trust artifacts and enable explicit consideration of uncertainty in decision-making processes related to reserve allocation and load management.

3.3. Forecast Explainability (XAI). The explainability module implements a multi-stage interpretation process, including the computation of SHAP attributions, grouping of contributing factors according to ontological classes, and generation of natural language explanations. This approach ensures logical and physical coherence of explanations: operators receive not merely technical indicators, but domain-oriented interpretations linked to specific wind or solar regimes, equipment states, or anomalous weather conditions. This closes the semantic gap and provides functional trust in terms of interpretability.

4. Model Management Layer: Drift, MLOps, and Lifecycle Management (DRIFT, OPS, ARCH). The upper layer of the architecture ensures operational trust and is responsible for maintaining correct model behavior over extended periods of operation.

4.1. Drift Monitoring (DRIFT). The drift monitoring module detects changes in input data distributions, variations in model error statistics, degradation of forecasting quality, and semantic inconsistencies associated with emerging regimes. It generates drift alerts and forwards them to the model management module for further action.

4.2. MLOps (OPS). The MLOps module provides automation for model training, CI/CD pipelines, service monitoring, container management, logging, availability control, and operational alerting. This component supports real-time reproducibility and stability of the forecasting system.

4.3. Model Lifecycle Management (OPS, ARCH). This component manages model storage within a Model Registry, records versions, metrics, and parameters, verifies model integrity, implements validation gates (decision gates), and supports safe deployment strategies (e.g., shadow deployment followed by rollout). It connects technological and functional trust by ensuring transparency, controllability, and compliance with regulatory requirements.

4.3. Overall Architectural Workflow

All eight modules form a unified end-to-end pipeline: data ingestion and semantic harmonization → data cleaning and feature construction → regime-aware forecasting → uncertainty estimation and explanation generation → drift monitoring → model support via MLOps and lifecycle management.

Thus, the proposed architecture represents not a collection of isolated algorithms, but a production-grade trustworthy forecasting system capable of maintaining robustness and interpretability throughout the entire operational lifetime of renewable energy generation assets.

The semantic layer constitutes a central element of the trust-oriented architecture. The underlying ontological model formalizes domain entities (e.g., wind profiles, solar irradiance, equipment states), their attributes, units of measurement, relationships, and admissible constraints. As a result, the semantic layer enables automatic validation of structural inconsistencies (such as unit or type mismatches), enforces feature consistency, ensures alignment of heterogeneous SCADA and NWP data sources, supports versioning of data structures, and provides interpretable representations for downstream modules. In this sense, semantics acts as a cross-cutting backbone of the architecture, establishing a robust foundation for explainability, adaptation, and operational reliability.

The semantic layer does not perform model interpretation itself but provides a consistent domain framework that enables explainability, uncertainty analysis, and governance mechanisms across the architecture.

Within the proposed architecture, an internal ontological model supports the semantic specification of features for renewable energy forecasting systems. The ontological layer is introduced as a structural mechanism for ensuring semantic consistency, interpretability, and quality control across heterogeneous data sources and processing stages. To provide an illustration of this approach, a wind power–specific ontology fragment is presented in Figure 1, serving as an illustrative example of how semantic modeling is applied within the architecture rather than as a universal or exhaustive domain ontology.

The ontological model is formalized using Web Ontology Language (OWL), which enables explicit representation of domain entities and their semantic relationships in a machine-interpretable form. The illustrative ontology fragment shown in Figure 4 captures relationships between sensor-level observations, meteorological variables, derived physical quantities, turbine characteristics, and electrical energy generation. In particular, classes such as HorizontalWindSpeed, VerticalWindSpeed, WindDirection, AirTemperature, AirHumidity, and AtmosphericPressure represent measured environmental variables, while derived concepts such as AirDensity, WindFlowPower, and TurbineEfficiency encode physically meaningful transformations relevant to wind energy conversion.

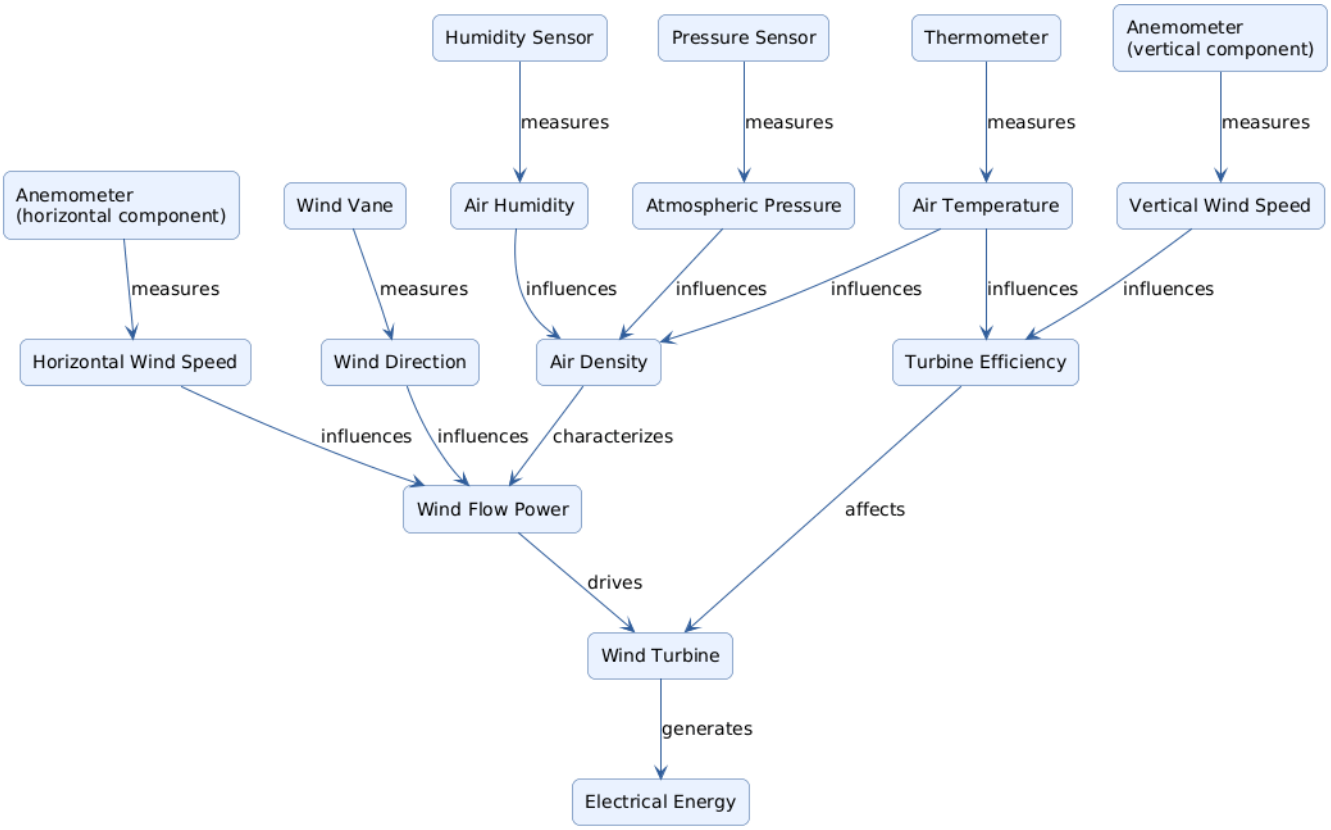


Figure 4. Illustrative ontology fragment for wind power forecasting, showing relationships between sensor measurements, meteorological variables, derived physical concepts, turbine characteristics, and electrical energy generation.

For each ontology class, a set of semantically defined attributes is specified, ensuring a consistent interpretation of physical quantities across the entire forecasting pipeline. Relationships between entities explicitly encode physical dependencies and operational logic, such as the influence of atmospheric conditions on air density, the characterization of wind flow power, and the conversion of wind energy into electrical output by the wind turbine.

These relationships reflect the underlying physical processes governing wind power generation and enable transparent reasoning about the factors affecting power output. During the preprocessing stage, the ontology is used as a declarative specification of the feature schema, enabling semantic validation of incoming data, detection of structural inconsistencies, and construction of interpretable features. A semantic mapping mechanism links input data fields to ontology entities, allowing automated and consistent feature extraction and supporting downstream tasks such as explainability, robustness analysis, and drift monitoring. While the example in Figure 4 focuses on wind power forecasting, the same OWL-based approach can be adapted to other renewable energy assets by extending or modifying the set of ontology classes and relationships according to asset-specific characteristics, available data sources, and operational requirements.

To explicitly link the structural gaps identified in Section 3.7 with the proposed architectural solution, Figure 5 illustrates the mapping between the identified gaps and the corresponding architectural components that address them.

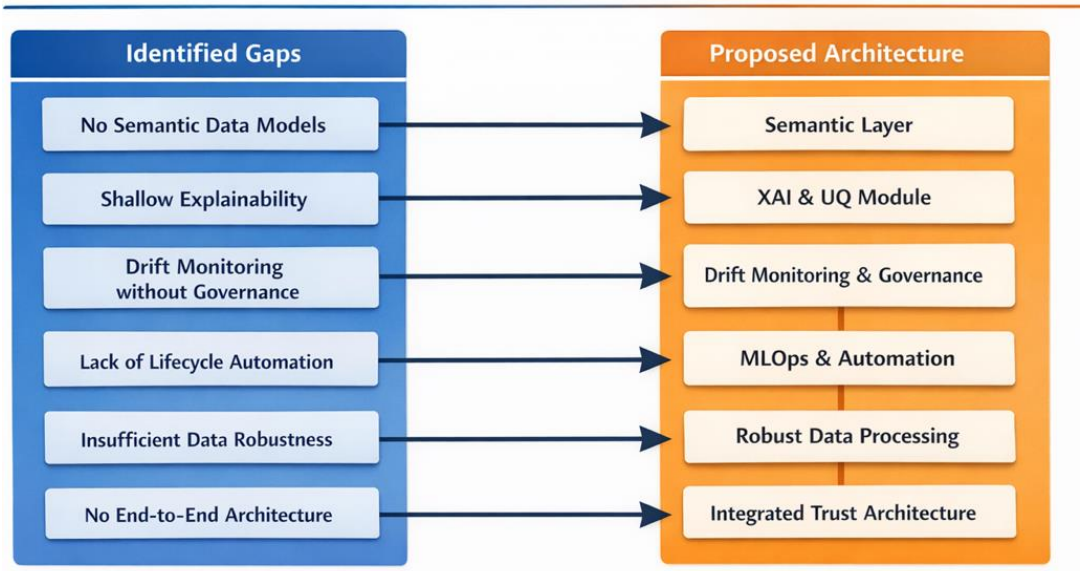


Figure 5. Mapping between the structural gaps identified in the literature review and the corresponding components of the proposed trustworthy forecasting architecture.

Table 9 shows the mapping between architecture's modules and trust attributes.

Table 9. Architecture's Modules and Trust Attributes.

Module	Trust Attribute	Metrics/Artifacts	Purpose
Input Data Validation and Harmonization	ARCH	Format and unit validation, data harmonization logs	Architectural integrity and correctness of the input data stream
Data Cleaning (Data QC)	ROB	Data quality (DQ) report, percentage of missing values, percentage of outliers, number of recovered data points	Robustness to noise and data defects
Feature Extraction and Regime Identification (Feature Layer)	ARCH, ROB	Feature map, stability report of identified regimes	Semantic consistency and reduction in intra-model variance
Forecasting Models	ARCH, ROB	MAE, RMSE, Nash–Sutcliffe Efficiency (NSE), residual distribution	Forecast accuracy and robustness

Table 9. *Cont.*

Module	Trust Attribute	Metrics/Artifacts	Purpose
Uncertainty Quantification (UQ)	UQ	CRPS, coverage, PIT, prediction interval width	Quantitative trustworthiness and calibration
Forecast Explainability (XAI)	XAI	Faithfulness score, stability score, textual explanations	Interpretability and transparency
Drift Monitoring	DRIFT	Drift alarms, distribution comparison metrics, degradation of performance metrics	Adaptivity and timely detection of degradation
MLOps and Model Lifecycle Management	OPS, ARCH	Model Registry, model versions, Model Cards, audit logs	Operational trust, manageability, and reproducibility

5. Discussion

The systematic review of contemporary studies on trustworthy artificial intelligence for wind and photovoltaic power forecasting, together with the proposed high-level architecture, provides an integrated perspective on how previously fragmented trust-related components can be combined into a single operationally viable framework. Unlike most existing works that focus on individual aspects—such as explainability, probabilistic forecasting, drift control, or robustness to data imperfections—this study highlights the necessity of their system-level integration within an architecture capable of supporting the full lifecycle of forecasting models.

5.1. Synthesis of Review Findings

The results of the systematic analysis indicate that current research on wind and photovoltaic power forecasting predominantly concentrates on individual trust attributes, including explainable AI, uncertainty quantification, data and model drift monitoring, robustness, as well as elements of MLOps and architectural pipelines. These research directions demonstrate substantial methodological progress; however, in most cases they are implemented in isolation, without forming a coherent and integrated system structure.

The literature analysis further shows that trust attributes are typically introduced as local extensions to specific models or isolated stages of the forecasting pipeline. Explainability, uncertainty estimation, robustness, and degradation monitoring are rarely treated as interdependent components of a unified architecture and, consequently, are weakly integrated with operational processes and model lifecycle management. As a result, existing solutions remain of limited applicability in real-world energy systems, where reproducibility, controllability, interpretability, and long-term stability of forecasting performance are critical.

This observed fragmentation confirms the presence of structural limitations at the architectural level of forecasting systems rather than at the level of individual algorithms or methods. These limitations were systematized as six key gaps in Section 3.7 and are subsequently used in this study as design requirements for synthesizing the integrated trustworthy forecasting architecture presented in Section 4.

5.2. Analysis of Research Trends

An analysis of publications from 2019 to 2025 reveals several persistent technological and methodological trends shaping the development of trustworthy wind and photovoltaic power forecasting systems.

The first trend is a shift in focus from model accuracy toward system-level trustworthiness. While earlier studies primarily emphasized improving forecasting accuracy through

ML/DL models, since approximately 2021–2022 there has been increasing attention to interpretability, probabilistic calibration, robustness to data imperfections, and monitoring of model degradation. Consequently, research emphasis has shifted from the "model as an algorithm" to the "model as a component of an architecture," increasing the importance of structural and operational considerations.

The second trend is the growing integration of explainability methods (XAI) into forecasting pipelines. Whereas early studies applied XAI mainly as a post hoc analysis tool, more recent works attempt to structure explanations, relate them to physical characteristics of the system, and integrate them into model validation procedures. Nevertheless, most such approaches remain algorithm-centric and lack semantic support, reinforcing the need for the ontology-driven approach proposed in this work.

The third trend is the expanding adoption of probabilistic forecasting and uncertainty quantification methods. Techniques such as quantile regression, ensemble dispersion, EMOS calibration, and conformal prediction are becoming increasingly common. However, the systematic integration of uncertainty quantification into architectural pipelines—particularly with respect to calibration requirements, reporting, and decision support—remains an open challenge.

The fourth trend is the increasing interest in drift monitoring and model adaptation. Since 2022, a growing number of studies emphasize the importance of continuous monitoring of data drift and model degradation. However, most solutions remain at the level of experimental prototypes and do not incorporate formalized model update procedures, underscoring the relevance of the operational management layer proposed in this study.

The fifth trend concerns the early-stage integration of MLOps practices into energy forecasting platforms. Although MLOps has become a standard paradigm in the broader AI industry, its adoption in the energy sector remains limited. While there is a clear movement toward automation, containerization, model repositories, and CI/CD pipelines, existing studies are still fragmented, and unified architectural solutions are largely absent, highlighting the practical relevance of the proposed architecture.

The sixth trend reflects a gradual transition from model-centric to architecture-centric perspectives. Recent studies increasingly recognize the need to view wind and photovoltaic power forecasting as a complex system comprising data, features, explanations, uncertainty handling, monitoring mechanisms, and operational infrastructure. Nevertheless, the literature still lacks comprehensive architectures that integrate all these elements, which emphasizes the significance of the contribution presented in this work.

Taken together, these trends indicate that wind and photovoltaic power forecasting is undergoing a transition from fragmented methodological solutions toward integrated architectural frameworks, with trustworthiness emerging as a key driving factor of this transformation.

5.3. How the Proposed Architecture Addresses the Identified Gaps

The proposed architecture provides a structural response to the gaps identified in Section 3.7 by integrating trust attributes at the level of the entire forecasting system rather than at the level of individual models or methods. Unlike existing solutions, the architecture treats trustworthiness as a system-level property, ensured through the coordinated operation of semantic, technological, and operational components.

First, the architecture is designed to address the gap related to the lack of semantic consistency in data and features by introducing an ontological layer that can be utilized across all stages of the pipeline—from the integration of SCADA and meteorological data to feature construction and explanation generation. This layer provides a structured mechanism for achieving a unified interpretation of physical quantities, operational regimes,

and system states, and is intended to improve reproducibility and extensibility of the forecasting system.

Second, the architecture introduces a semantically grounded XAI approach intended to overcome the limitations of shallow model-centric explainability. Within this design, model attributions can be interpreted in terms of domain-relevant factors, enabling traceable explanations across the forecasting pipeline and supporting engineering-oriented interpretation rather than purely post hoc analysis of model behavior.

Third, the architecture is designed to bridge the gap between drift detection and model lifecycle management by integrating monitoring of data degradation and forecasting performance with formalized procedures for adaptation, validation, and safe model updates. This allows drift to be treated not as an experimental artifact, but as a potentially manageable operational risk.

Fourth, through the integration of data quality control modules, ontology-driven preprocessing, and regime clustering, the architecture is intended to enhance robustness to missing values, noise, and structural inconsistencies in data that are characteristic of real SCADA and NWP streams. Robustness is thus addressed not at the level of individual algorithms, but as a design-level property of the overall architecture.

Finally, the architecture incorporates operational aspects of trustworthiness through the integration of MLOps and model governance. Version control, validation gates, artifact auditing, rollback mechanisms, and controlled deployment strategies together form a transparent and manageable model lifecycle, which fundamentally distinguishes the proposed approach from the majority of existing frameworks.

In summary, the proposed architecture serves as an integrated end-to-end reference model that addresses the identified structural gaps and ensures semantic, technological, and functional trustworthiness of wind and photovoltaic power forecasting systems under real operational conditions.

5.4. Comparison of the Proposed Architecture with Existing Frameworks

A comparison of the proposed architecture with existing frameworks for trustworthy renewable energy forecasting indicates that most published solutions are oriented toward addressing individual sub-tasks and rarely form a coherent architecture that spans the full lifecycle of a forecasting system.

Frameworks primarily focused on data quality and robustness to missing values and noise typically incorporate data quality–aware layers and input stream monitoring procedures. However, such solutions seldom integrate uncertainty quantification, formalized explainability mechanisms, and model lifecycle management within a single architectural framework. Similarly, approaches centered on explainability most often implement XAI at the level of individual models (e.g., through SHAP or LIME), without relying on a semantic domain model or ensuring coherence between explanations, physical system components, and operational regimes.

Some studies propose more comprehensive pipelines that account for data drift or incorporate elements of energy-oriented MLOps pipelines, including automation of training, deployment, and model monitoring. Nevertheless, even these solutions tend to remain fragmented: the semantic layer is used in a limited manner or is absent altogether; XAI and uncertainty quantification are not treated as mandatory architectural components; and the relationships between trust control points and system modules are left informal.

In contrast to these approaches, the proposed architecture is positioned as an integrated reference model characterized by the following properties:

- Coverage of the entire forecasting system lifecycle, from data integration and preprocessing to forecasting, uncertainty quantification, explainability, and operational support;

- Use of an ontological semantic layer as a cross-cutting mechanism for aligning data, features, models, and explanations;
- Integration of key trust attributes (ARCH, ROB, UQ, XAI, DRIFT, OPS) within a unified structured pipeline, with explicit traceability of modules and artifacts;
- Architectural agnosticism with respect to the type of generation, while remaining applicable to both wind and photovoltaic power plants through differences at the level of features and regimes rather than architectural principles.

Thus, the proposed architecture does not oppose existing frameworks at the level of individual methods; rather, it generalizes and systematizes their strengths in the form of a holistic end-to-end trustworthy forecasting model. The presence of formalized trust attributes and measurable artifacts (see the traceability table) enhances the applicability of the architecture under real operational conditions of wind and photovoltaic power plants.

Beyond architectural comparison, an important question concerns the practical implications of integrating trust attributes into renewable energy forecasting systems. While the proposed architecture is formulated at a conceptual level, its design directly addresses operational challenges faced by wind and photovoltaic power plants. These implications are discussed in the following subsection.

5.5. Practical Implications of Integrating Trustworthy AI for Renewable Energy Applications

The integration of trust attributes within a unified architectural framework leads to several practical effects that are directly relevant to the operation of wind and photovoltaic power plants:

- Reduction in operational risks through calibrated forecasts and improved uncertainty handling in reserve capacity planning;
- Increased acceptance of AI-based decisions by engineers and operators, as explanations become transparent, structured, and physically interpretable;
- Stabilization of forecasting performance through timely detection and correction of data and model drift;
- Enhanced robustness to input data defects, which is particularly critical when working with real-world SCADA data streams;
- Scalability and portability, as the architecture is unified for wind and photovoltaic assets and adaptable to hybrid energy systems. trustworthy AI; wind power forecasting; photovoltaic forecasting; explainable AI; uncertainty quantification; MLOps; drift monitoring; semantic modeling.

These effects position the proposed architecture not only as a research concept but also as a practical foundation for industrial deployment of trustworthy AI-based forecasting systems in renewable energy applications.

5.6. Study Limitations and Threats to Validity

Despite the comprehensive nature of the conducted analysis, the study is subject to several limitations related to the literature selection methodology, the completeness of domain coverage, and the interpretation of results. These factors do not diminish the significance of the findings but should be considered when generalizing the results and applying the proposed architecture in practice.

1. Internal validity. The literature selection was performed using several major databases (Scopus, IEEE Xplore, MDPI, and ScienceDirect); however, the completeness of the sample is constrained by the applied search queries and the availability of abstracts. Relevant studies employing alternative terminology or lacking explicit references to trust attributes may have been omitted. In addition, some studies may implicitly

incorporate trust-related properties without explicitly highlighting them, which could lead to incomplete interpretation.

2. External validity. The results primarily reflect the state of research reported in academic literature and, to a lesser extent, in engineering publications. Industrial solutions, internal developments within companies, and commercial wind and photovoltaic forecasting platforms are not always publicly documented and were therefore not included in the analysis. Consequently, the external validity of the findings is limited by the lack of extensive evidence on the deployment of trustworthy architectures in fully operational power systems.
3. Construct validity. The assessment of trust attributes was based on textual and structural analysis of the reviewed publications. The actual degree of implementation of each attribute may differ from what is reported, and the depth of implementation (e.g., the completeness of explainability or the quality of uncertainty calibration) may vary across studies. Moreover, not all XAI, UQ, or drift detection methods are associated with standardized evaluation metrics, which complicates direct comparison between approaches.
4. Selection bias. Despite adherence to the PRISMA methodology, the selected corpus may exhibit a bias toward studies focused on ML/DL-based models, while traditional physics-based models, hybrid approaches, and industrial architectures are less represented. In addition, some publications were excluded due to insufficient architectural or methodological descriptions, which may further contribute to potential selection bias. A formal risk-of-bias assessment using standardized tools was not performed, as the included studies do not represent interventional or comparative effect studies. The review focuses on architectural and methodological characteristics rather than quantitative outcome estimation.

5.7. Limitations of the Architecture

Despite the outlined practical benefits, the proposed architecture also entails several limitations that should be explicitly acknowledged when considering real-world deployment.

First, the architecture is currently presented at the level of conceptual design; its validation requires implementation across heterogeneous systems and geographic regions. Second, full-scale realization presupposes the availability of a mature MLOps infrastructure, which is still lacking at many energy facilities. Third, the semantic explainability module requires additional evaluation involving engineers and dispatch operators to assess its practical effectiveness. Finally, several components—such as ontological quality control, regime clustering, and explanation generation—require adaptation to specific IT stacks and data sources.

At the current stage, the proposed architecture is validated at a conceptual and design level. Comprehensive experimental validation on real-world datasets and operational systems is beyond the scope of this study and is considered an important direction for future work.

5.8. Answers to Research Questions

Answer to RQ1: Which trust attributes are represented in current studies on wind and photovoltaic power forecasting?

The analysis of a corpus of 52 publications (PRISMA 2020) indicates that contemporary studies incorporate individual trust attributes, but do so in an imbalanced and fragmented manner. The most frequently reported attributes are:

- XAI (post hoc explainability, SHAP/LIME): 26 studies;
- Robustness (resilience to missing data and noise): 25 studies;

- Drift monitoring: 21 studies;
- Uncertainty quantification (UQ): 13 studies;
- MLOps/operationalization: 12 studies;
- Model governance: 7 studies;
- Architectural frameworks or pipelines: only isolated cases.

Notably, none of the reviewed publications simultaneously incorporates all trust attributes, and none proposes a fully integrated end-to-end architecture.

Answer to RQ2: What gaps and unresolved problems exist in contemporary architectures?

The review reveals six structural gaps that prevent the development of industrial-grade trustworthy renewable energy forecasting systems: lack of semantic data models, shallow model-centric explainability, drift monitoring without governance mechanisms, absence of automated lifecycle management, insufficient robustness to real operational data defects, and the lack of end-to-end architectures. A detailed analysis of these gaps is provided in Section 3.7.

Answer to RQ3: What requirements must a trustworthy wind and photovoltaic forecasting system satisfy?

The identified gaps lead to the formulation of the following requirements:

Semantic requirements:

- Data ontology;
- Unified feature schema;
- Verifiability and reproducibility of data.
- Technological requirements:
- Robustness to all types of data defects (missing values, noise, outliers, synchronization errors);
- Availability of regime-specific models;
- Uncertainty estimation and calibration;
- Drift monitoring with actionable response mechanisms.
- Functional requirements:
- Transparency, explainability, and human-readable interpretations;
- Formalized model lifecycle management;
- CI/CD pipelines, model registries, and audit trails;
- Mechanisms for safe model updates and rollback.

Answer to RQ4: How can the identified elements be synthesized into a unified trustworthy AI forecasting architecture?

The synthesis is based on a four-layer architectural concept, in which each layer implements a specific set of trust attributes:

5. Semantic Layer (ARCH)—data harmonization, ontology, and validation of units and structures;
6. Data and Feature Layer (ROB, ARCH)—data cleaning, feature construction, and regime clustering;
7. Modeling, UQ, and XAI Layer (ARCH, ROB, UQ, XAI)—forecasting, probabilistic estimation, and interpretation;
8. Operational Trust Layer (DRIFT, OPS, ARCH)—drift monitoring, MLOps practices, and model lifecycle management.

6. Conclusions

This study presented a systematic analysis of contemporary approaches to the development of trustworthy artificial intelligence systems for wind and photovoltaic power forecasting, conducted in accordance with the PRISMA methodology. The review of rel-

evant publications demonstrated that, despite the active development of explainability methods, probabilistic forecasting, drift monitoring, and robustness mechanisms, existing solutions largely remain fragmented at the architectural level and insufficiently aligned with the operational requirements of real-world energy assets.

Based on the identified structural limitations, a high-level architecture of a trustworthy forecasting system for wind and photovoltaic power plants was proposed. The architecture formalizes trustworthiness as a system-level property and integrates semantic, technological, and functional layers within a unified data and model lifecycle—from harmonization of heterogeneous data sources and quality control to forecasting, uncertainty quantification, explainability, degradation monitoring, and controlled model adaptation.

The proposed architectural approach shifts the focus from improving individual algorithms toward the development of reproducible, interpretable, and manageable forecasting systems. By treating explainability, robustness, uncertainty, and operational reliability as interdependent components, the architecture can serve as a reference foundation for the design and deployment of industrially oriented renewable energy forecasting solutions.

As the primary contribution of this work lies in the systematic synthesis of trust attributes and the design of a reference architecture, the focus is placed on architectural consistency and conceptual soundness rather than empirical performance evaluation.

Future research may focus on experimental validation of the proposed architecture across different types of renewable energy installations, quantitative assessment of the contribution of individual trust layers to forecasting stability, and formalization of trust requirements in accordance with existing and emerging industry standards. An additional direction involves the integration of the proposed architecture into digital twins of energy systems and multi-asset generation complexes.

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Abbreviations

The following abbreviations are used in this manuscript:

ARCH	Architectural Trust
UQ	Uncertainty Quantification Trust
DRIFT	Drift Monitoring and Adaptation Trust
OPS	Operational Trust (MLOps and Governance)
ROB	Robustness Trust
AI	Artificial Intelligence
ARIMA	Autoregressive Integrated Moving Average
AutoML	Automated Machine Learning
BEM	Blade Element Momentum
CI/CD	Continuous Integration and Continuous Deployment

CNN	Convolutional Neural Network
CRPS	Continuous Ranked Probability Score
DQ	Data Quality
ECE	Expected Calibration Error
EMOS	Ensemble Model Output Statistics
EMS	Enterprise Management System
ERP	Enterprise Resource Planning
FL	Federated Learning
GRU	Gated Recurrent Units
HBOLA	Hedge Backpropagation-based Online LSTM Architecture
LIME	Local Interpretable Model-agnostic Explanations
LSTM	Long Shot-term Memory
MAE	Mean Absolute Error
MLOps	Machine Learning Operations
NSE	Nash–Sutcliffe Efficiency
NWP	Numerical Weather Prediction
ONE-NAS	Online NeuroEvolution-based Neural Architecture Search
OSF	Open Science Framework
OWL	Web Ontology Language
PDF	Probability Density Function
PIT	Probability Integral Transform
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
RES	Renewable Energy Sources
RMSE	Root Mean Square Error
RNN	Recurrent Neural Networks
SCADA	Supervisory Control and Data Acquisition
SHAP	Shapley Additive exPlanations
SLR	Systematic Literature Review
UQ	Uncertainty Quantification
XAI	Explainable Artificial Intelligence

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